MVP DEMO PRESENTATION

Presented by : Team 16

Our Team



Jessica goel

Frontend Developer



Mridul Nohria

Integration Lead / DevOps / QA Lead



Yerdana Maulenbay

Backend Developer



Rhythm Trivedi

Frontend Developer

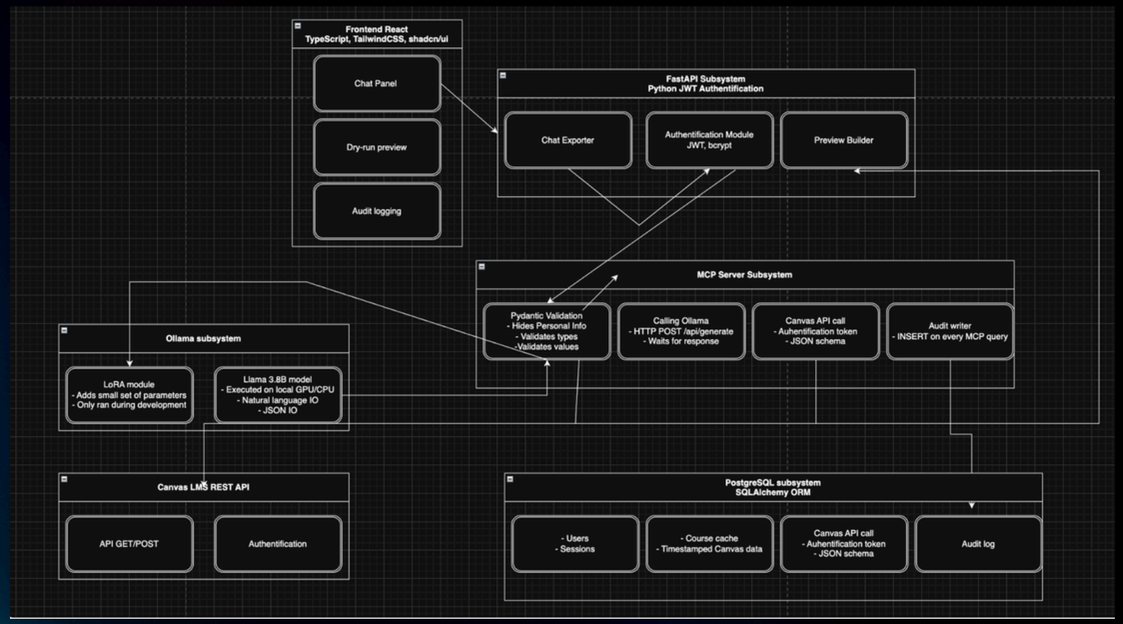


Bao Tran

Backend Developer

CHANGES MADE

OLD SYSTEM ARCHITECTURE



- Single local Ollama model for AI processing.
- AI requests routed directly through the MCP server.
- Monolithic backend with fewer separated services.
- Basic database structure for authentication and Canvas data.

NEW SYSTEM ARCHITECTURE



- Added a provider-independent LLM Adapter Layer supporting both local and cloud-hosted models.
- Refactored the backend into modular routers (Authentication, Canvas, Chat, Preview, Messages, Admin).
- Introduced a workflow engine with validation before Canvas actions.
- Expanded PostgreSQL to support conversations, workflows, templates, and audit logs.
- Improved scalability, maintainability, and AI safety.

WHY WE CHANGED IT

- Support multiple LLM providers instead of relying only on local models.
- Reduce hardware requirements by allowing cloud-hosted models.
- Make the system easier to extend and maintain.
- Improve security and validation before executing Canvas operations.
- Better align the architecture with project requirements and future milestones.

Impact: The new architecture is more scalable, supports multiple AI providers, improves security, and simplifies future feature development.

COMPLETED TASKS TESTING REPORT

	REQUIREMENTS	TYPE OF TESTING(I = Integration Testing S = System Functionality Testing O = Operational Acceptance Testing UN = Unit Testing US = Usability Testing A = Acceptance Testing,- = not applied)	PASS OR FAIL	CONTRIBUTOR	STATUS
	USER REQUIREMENTS				
	INSTRUCTOR				
1.1	Must log in securely and access administrative functions	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
1.2	Must be able to create, update, enable, disable, and manage instructor accounts	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
1.3	Must be able to revoke instructor access when required.	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
	ADMINISTRATOR				
2.1	Must be able to securely create account, authenticate and access the application	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.2	Must be able to securely connect a Canvas API token.	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
2.3	Must be able to replace or remove a stored Canvas API token.	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
2.4	Must be able to view Canvas courses associated with the connected token.	UN,I,S	PASS	RHYTHM,MRIDUL,BAO	☑
2.5	Must be able to select an active course.	UN,I,S	PASS	RHYTHM,MRIDUL,BAO	☑
2.6	Must be able to view course information, students, assignments, and submissions.	UN,I,S	PASS	JESSICA,MRIDUL,BAO	☑
2.7	Must be able to analyze missing submissions.	UN,I,S	PASS	RHYTHM,MRIDUL,BAO	☑
2.8	Must be able to analyze late submissions.	UN,I,S	PASS	RHYTHM,MRIDUL,BAO	☑
2.10	Must be able to access a dashboard summarizing course activity.	UN,I,S	PASS	RHYTHM,MRIDUL,BAO	☑
2.11	Must be able to interact with the system using natural-language requests.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.12	Must be able to ask questions about course data.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.13	Must be able to receive explanations of proposed actions before execution.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.14	Must be able to create Canvas assignments.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.15	Must be able to update assignment details and dates.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.18	Must be able to draft Canvas messages to selected students.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.19	Must be able to review and edit messages before sending.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.23	Must be able to preview Canvas-modifying actions before execution.	UN,I,S	PASS	JESSICA,MRIDUL,YERDANA	☑
2.24	Must explicitly approve Canvas-modifying actions before execution.	UN,I,S	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.25	Must be able to cancel a proposed action before execution.	UN,I,S	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.26	Must be able to view a history of previous actions.	UN,I,S	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.27	Must be able to view previous assistant requests and their outcomes.	UN,I,S	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.29	Must receive clear feedback when requests cannot be completed safely.	UN,I,S	PASS	RHYTHM,MRIDUL,YERDANA	☑

BURN UP



Reflection (Up to Today)

- Project scope expanded significantly as detailed MVP features and sub-issues were added during planning.
- Steady completion of frontend, backend, and integration features throughout the milestone.
- Large increase in completed work near the MVP deadline as integrated features were finalized and merged.

Moving Forward

- Continue completing remaining milestones with more consistent progress each week.
- Increase focus on testing, bug fixing, and code quality.
- Maintain smaller, incremental issue completion to reduce end-of-milestone workload.

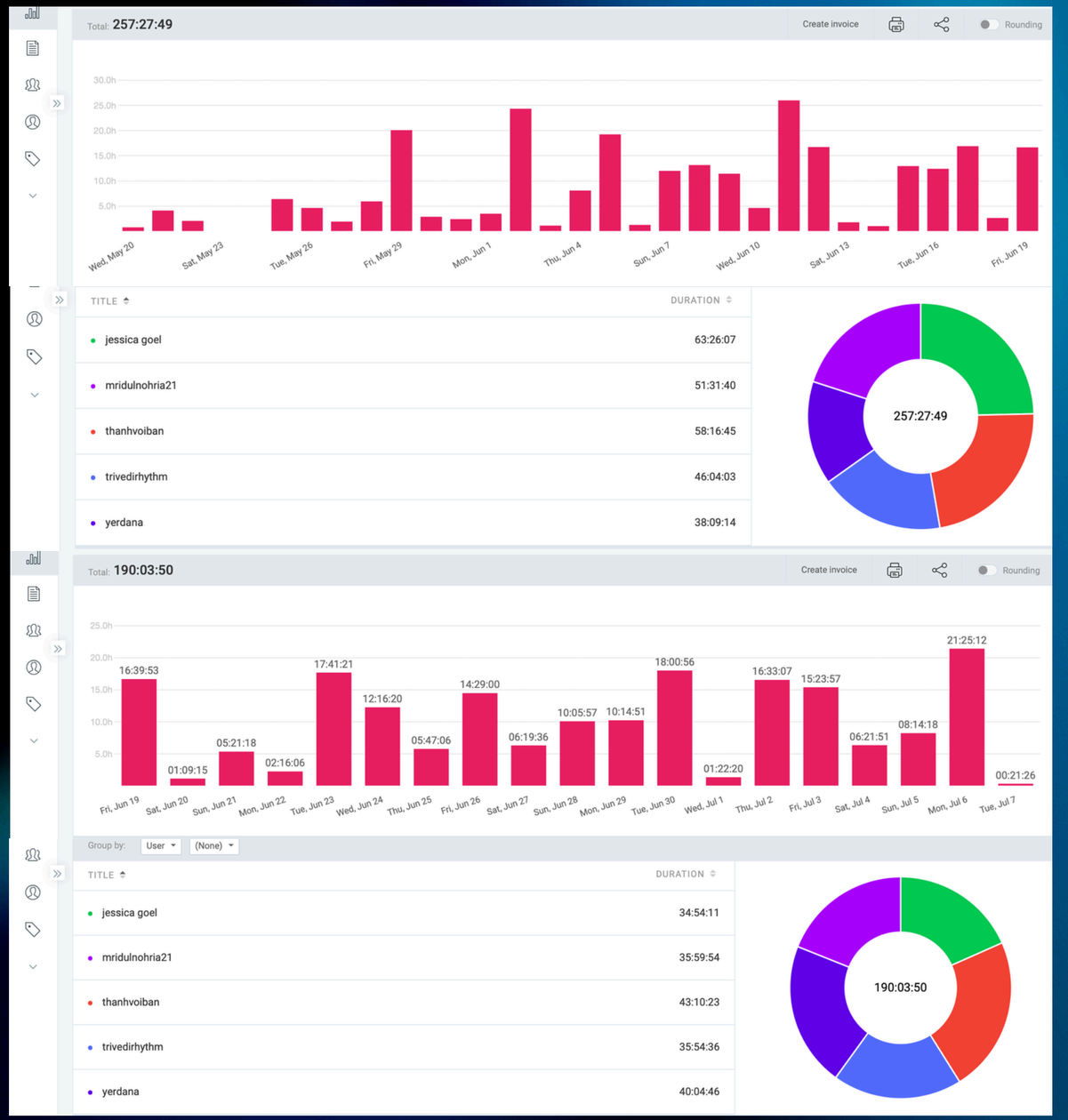
TIME CONTRIBUTION

Progress to Date

- 448 team hours logged.
- Completed all planned MVP frontend, backend, and integration features.
- Successfully delivered Feature Set #1 and achieved MVP functionality.
- Development effort focused on implementation, integration, and validation

Moving Forward

- Prioritize comprehensive testing and bug fixing.
- Improve test coverage and system reliability.
- Refine UI/UX based on feedback.
- Continue implementing remaining features for the final release.



TESTING

Testing Type	Framework/Tool	Coverage	Status
Unit Tests (Models/Logic)	Pytest	User, Course, Application models	Active
Frontend Tests	Vitest	DOM behavior form validation	Active
CI Test Automation	GitHub Actions	Run on every PR	Active

FRONTEND TESTS:

```
Problems  Output  Debug Console  Terminal  Ports  Azure  Query Results

 ✓ lib/validation.test.ts (6 tests) 9ms
 ✓ pages/Instructor/InstructorSubmissionsAnalysis/submissionUtils.test.ts (5 tests) 8ms
 ✓ lib/audit-log-users.test.ts (1 test) 13ms
 ✓ schemas/instructorSchema.test.ts (3 tests) 3ms
 ✓ services/auditLogs.test.ts (3 tests) 9ms
 ✓ services/workflowHistory.test.ts (5 tests) 10ms
 ✓ services/canvas.test.ts (6 tests) 28ms
 ✓ services/assignmentCreator.test.ts (3 tests) 3ms
 ✓ services/instructorAccounts.test.ts (2 tests) 7ms
 ✓ services/messageDraft.test.ts (4 tests) 4ms
 ✓ lib/canvas-url.test.ts (6 tests) 7ms
 ✓ lib/analytics-trends.test.ts (7 tests) 4ms
 ✓ lib/auth-errors.test.ts (12 tests) 6ms
 ✓ lib/map-audit-log.test.ts (4 tests) 8ms
 ✓ lib/onboarding-storage.test.ts (2 tests) 2ms
 ✓ pages/Instructor/InstructorDashboard/dashboardColors.test.ts (2 tests) 3ms
 ✓ lib/sidebar-storage.test.ts (2 tests) 3ms
 ✓ lib/table-pagination.test.ts (3 tests) 8ms
 ✓ data/mockInstructors.test.ts (1 test) 3ms
 ✓ data/institutionSubmissionsData.test.ts (2 tests) 2ms
 ✓ data/mockDryRunPreview.test.ts (3 tests) 4ms
 ✓ lib/map-workflow-history.test.ts (3 tests) 3ms
 ✓ lib/institution-storage.test.ts (2 tests) 1ms
 ✓ lib/workflow-history-query.test.ts (7 tests) 6ms
 ✓ data/mockAuditLogs.test.ts (5 tests) 4ms
 ✓ pages/Instructor/InstructorAssignmentCreate/assignmentCreateSchema.test.ts (4 tests) 30ms
 ✓ lib/canvas-connection.test.ts (4 tests) 4ms
 ✓ lib/assignment-utils.test.ts (3 tests) 4ms
 ✓ lib/map-intent-preview-to-dry-run.test.ts (1 test) 3ms
 ✓ pages/Instructor/InstructorSubmissionsAnalysis/submissionExport.test.ts (2 tests) 1ms
 ✓ lib/format-course-title.test.ts (3 tests) 2ms
 ✓ lib/institution-canvas.test.ts (2 tests) 1ms
 ✓ lib/user-display.test.ts (3 tests) 1ms
 ✓ lib/format-audit-timestamp.test.ts (2 tests) 1ms
 ✓ lib/format-relative-time.test.ts (1 test) 1ms

 Test Files  143 passed (143)
      Tests  522 passed (522)
   Start at  06:38:48
   Duration  16.76s (transform 3.05s, setup 13.75s, collect 17.85s, tests 40.10s, environment 50.22s, prepare
8.41s)
```

BACKEND TESTS:

```
  /Users/jessicagoel/capstone-team-16-2/app/backend/.venv/lib/python3.14/site-packages/pytest_asyncio/plugin.p
y:869: DeprecationWarning: 'asyncio.get_event_loop_policy' is deprecated and slated for removal in Python 3.16
    policy = asyncio.get_event_loop_policy()

test_admin_auth.py: 12 warnings
test_admin_management.py: 9 warnings
test_admin_settings.py: 15 warnings
test_assignment.py: 34 warnings
test_auth.py: 28 warnings
test_canvas.py: 62 warnings
test_chat_cancel.py: 28 warnings
test_chat_confirm.py: 17 warnings
test_chat_endpoint.py: 14 warnings
test_course_overview.py: 9 warnings
test_error_handling.py: 4 warnings
test_history.py: 4 warnings
test_instructor_dashboard.py: 4 warnings
test_instructor_management.py: 47 warnings
test_late_submissions.py: 23 warnings
test_message_send_validation.py: 16 warnings
test_message_templates.py: 11 warnings
test_missing_submissions.py: 22 warnings
test_new_features.py: 8 warnings
test_permissions.py: 10 warnings
test_security.py: 7 warnings
test_student.py: 34 warnings
test_workflow_history.py: 22 warnings
  /Users/jessicagoel/capstone-team-16-2/app/backend/.venv/lib/python3.14/site-packages/jose/jwt.py:311: Deprec
ationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use time
zone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = timegm(datetime.utcnow().utctimetuple())

test_message_send_validation.py::test_send_rejects_recipient_not_enrolled
test_message_send_validation.py::test_send_rejects_recipient_when_second_of_two_not_enrolled
  /Users/jessicagoel/capstone-team-16-2/app/backend/app/services/message_validation.py:78: StarletteDeprecatio
nWarning: 'HTTP_422_UNPROCESSABLE_ENTITY' is deprecated. Use 'HTTP_422_UNPROCESSABLE_CONTENT' instead.
    await _validate_recipients_in_course(

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================== 313 passed, 1 skipped, 23207 warnings in 39.92s ==========================
```

LIVE DEMO

01 **Natural Language Assistant**

02 **Course Dashboard & Submission Analytics**

03 **Assignment Creation with Dry-Run Preview**

04 **Instructor Account Management**

ANY QUESTIONS?

THANK YOU